

The Solubility Curve of Ammonium Chloride

Purpose:

To review solution terminology and to investigate the relationship between temperature and the solubility of ammonium chloride in water

Terms:

You should be able to define the following terms:

solution

unsaturated solution

supersaturated solution

solution

Materials:

graduated pipette

thermometer

retort stand

stirring rod

test tube clamp

Bunsen burner

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Procedure:

1. Accurately mass approximately 7.00 g of Ammonium Chloride in a large test tube. Add to the test tube 10.0 mL of water using a graduated pipette. Support the test tube at an angle on a retort stand and insert a thermometer and a stirring rod.
2. Heat the contents to about 90 °C while stirring constantly. Do not boil the solution. Continue heating and stirring until all of the solid has dissolved.
3. Allow the solution to cool slowly, stirring gently as it cools. Watch it very closely. At the first sign of solid forming in the solution, note the temperature and record it in the table. This should be very close to the temperature where 7.00 g of Ammonium Chloride forms a saturated solution with 10.0 g of water (assuming water has a density of 1.0 g/mL).
4. Add 2.0 mL of water to the test tube, stir and warm (if necessary) to dissolve the Ammonium Chloride. Save time by warming only to the temperature where the crystals completely dissolve.
5. Again, let the solution cool while stirring. Again, record the temperature where crystallization first occurs. This should be very close to the temperature where 7.00 g of Ammonium Chloride forms a saturated solution with 12.0 g of water.
6. Add a second 2.0 mL portion of water and repeat the procedure.
7. Add a third 2.0 mL portion of water and repeat the procedure.
8. Add a fourth 2.0 mL portion of water and repeat the procedure.
9. Make a solubility graph of your results. Put Solubility (in g/100g of water) on the y axis and Temperature (in °C) on the x axis.

Observations:

Mass of NH_4Cl (g)	Mass of Water (g)	Concentration (g of solute/g of water)	Solubility (g of solute/100 g of water at saturation)	Temperature (°C)

Discussion Questions:

1. What generalization can you make regarding the effect of heat on the solubility of a solid?
2. What is the solubility of Ammonium Chloride at a) 20 °C ? B) 60 °C ?
3. At what temperature will a solution of 25 g of Ammonium Chloride in 100 g of water be saturated?
4. What mass of Ammonium Chloride is needed to make a saturated solution in 100 g of water at 40 °C?
5. If a saturated solution of Ammonium Chloride (in 100 g of water) at 50 °C is cooled to 20 °C, what mass of precipitate will be formed?
6. What would you expect the solubility curve of another substance to look like?